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|-----------|------------|---|----------------|-----|
| 11 | (a) | (induced) e.m.f. proportional/equal to <u>rate</u>
of change of (magnetic) flux (linkage) | M1
A1 | [2] |
| | (b) | (for same current) iron core gives large(r) (rates of change of) flux (linkage)
e.m.f induced in solenoid is greater (for same current)
induced e.m.f. opposes applied e.m.f. so current smaller/acts to reduce current | B1
M1
A1 | [3] |

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or

same supply so same induced e.m.f. balancing it
 (rate of change of) flux linkage is same
 smaller current for same flux when core present

(B1)
 (M1)
 (A1)

(c) e.g. (heating due to) eddy currents in core

(heating due to current in) resistance of coils

hysteresis losses/losses due to changing magnetic field in core

Any two of the above marking points, 1 mark each

B2 [2]